

From MOFs to Single Fe Atoms Implanted N-doped Porous Carbons: Efficient Oxygen Reduction in Both Alkaline and Acidic Media

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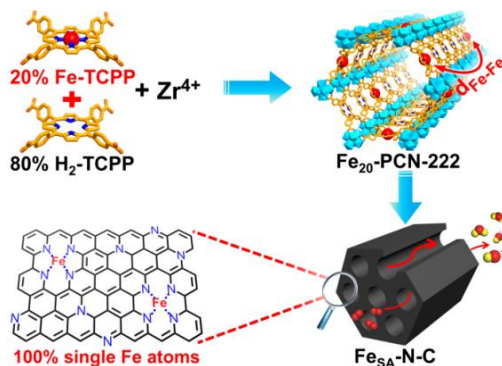
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It remains highly desired but a great challenge to achieve atomically dispersed metals in high loadings for efficient catalysis. Herein, for the first time, we have developed porphyrinic metal-organic frameworks (MOFs) based on a novel mixed-ligand strategy to afford high-content (1.76 wt%) single-atom (SA) Fe implanted N-doped porous carbon (FeSA-N-C) via pyrolysis. Thanks to the SA Fe sites, hierarchical pores, oriented mesochannels and high conductivity, the optimized FeSA-N-C exhibits excellent oxygen reduction activity and stability, surpassing almost all non-noble-metal catalysts and state-of-the-art Pt/C, in both alkaline and more challenging acidic media. More importantly, this MOF-based mixed-ligand strategy opens a novel avenue to the precise fabrication of efficient single-atom catalysts.

Single-atom catalysts (SACs), a new research frontier in heterogeneous catalysis, have been intensively investigated in diverse reactions, including electrocatalysis, photocatalysis, organic catalysis, etc.¹⁻⁴ Given that only the coordinatively unsaturated metal atoms on the surface of solid catalysts could contribute to the catalytic process, SACs would realize the utmost utilization of metal sites and thus greatly benefit the catalytic efficiency.² Furthermore, SACs are able to overcome the complicated multi-components in traditional supported-metal catalysts and regarded as ideal model catalysts to identify active sites and understand reaction process at a molecular level.^{2,3} Though great progress has been achieved, the rational fabrication of SAC with high metal loadings (>1 wt%) remains a grand challenge.⁴ In addition to the introduction of more active metals, the optimized pore structure, large surface area and excellent conductivity in SAC-based catalysts, which would facilitate mass/electron transfer, are also desirable to improve the activity.⁵

To meet these challenges, metal-organic frameworks (MOFs),⁶ a class of crystalline porous materials constructed by metal ions/clusters and versatile organic linkers, might be promising candidates. MOFs have been demonstrated to be great precursors/templates to produce porous carbon-based materials via pyrolysis.^{7,8} More importantly, their periodic structures give rise to spatial separation of building units and thus inhibit potential agglomeration of metal sites during pyrolysis, enabling MOFs ideal precursors to create SACs. Currently, the reports on MOF-pyrolyzed materials are mostly focused on the regulation of their pore features and compositions.^{7d-g,8} The precise fabrication of active sites at molecule level based on MOF precursors has been rarely reported. Very recently, zeolitic imidazolate frameworks (ZIFs) were reported as precursors to successfully produce SACs based on structural tailorability, high nitrogen content and pore confinement effect.^{4b,9} Unfortunately, the microporous feature of ZIFs and their derivatives may severely hamper mass transfer and the accessibility of active sites.^{9b} Therefore, it is imperative to develop SACs (providing efficient active sites) with hierarchical pores (benefitting mass transfer and active site accessibility) via rational design of MOF precursors for improved catalytic performance.

Scheme 1. Illustration of the rational fabrication of single Fe atoms-involved Fe_{SA}-N-C catalyst via a mixed-ligand strategy.



With the above in mind, a porphyrinic MOF, PCN-222 (also called MOF-545 or MMPF-6),¹⁰ featuring 1D mesochannels with a diameter of 3.2 nm (Scheme S1), was chosen as a representative precursor. The ratio modulation between Fe-TCPP (TCPP = tetrakis (4-carboxyphenyl)porphyrin) and H₂-TCPP mixed ligands affords a series of isostructural MOFs, denoted as Fe_x-PCN-222 (x%: molar percentage of Fe-TCPP in both ligands) (Scheme S2). The assembly of Fe-TCPP into 3D networks of Fe_x-PCN-222 effectively inhibits the molecular stacking, and the mixed-ligand strategy further elongates distance of adjacent Fe-TCPP ligands. Upon pyrolysis, the optimized Fe₂₀-PCN-222 can be converted into single-atom (SA) Fe sites implanted porous N-doped carbon, denoted as Fe_{SA}-N-C, with unique mesopore character (Scheme 1). By integration of single Fe atoms (highly active sites) and hierarchically porous structure with oriented/ordered mesopores (facilitating active site access and mass transfer), the optimized Fe_{SA}-N-C exhibits superb oxygen reduction activity and ultrahigh stability, surpassing almost all reported non-noble-metal catalysts and the state-of-the-art Pt/C, under both alkaline and acidic conditions.

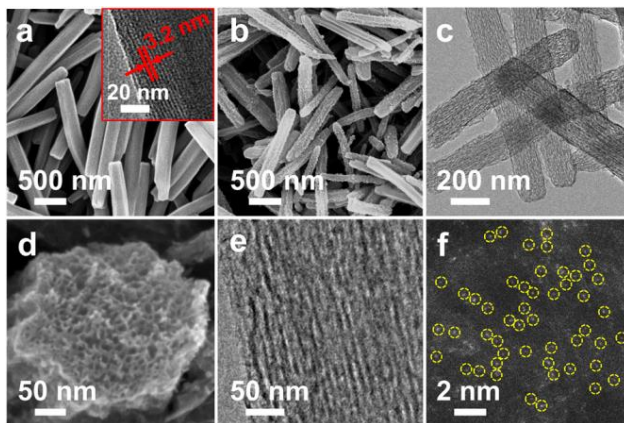


Figure 1. (a) Scanning electron microscopy (SEM) image of Fe₂₀-PCN-222 (inset: transmission electron microscopy (TEM) image showing the mesochannels in Fe₂₀-PCN-222). (b) SEM and (c) TEM images of Fe_{SA}-N-C. (d) Cross-section SEM and (e) enlarged TEM images of the mesoporous structure of Fe_{SA}-N-C. (f) Aberration-corrected HAADF-STEM image of Fe_{SA}-N-C.

A series of Fe_x-PCN-222, including Fe₀-PCN-222, Fe₂₀-PCN-222 and Fe₄₀-PCN-222, have been prepared and they present similarly high surface areas with uniform pore size distribution (Figure S1 and S2). Taking Fe₂₀-PCN-222 as a representative, its typical type-IV N₂ isotherms suggest a Brunauer-Emmett-Teller (BET) surface area as high as 2062 m²/g and two types of pores, with sizes of 1.2 nm and 3.2 nm, corresponding to the triangular microchannels and hexagonal mesochannels, respectively. SEM and TEM images present the uniform rod-shaped morphology with a diameter of ~200 nm (Figure 1a and S3a). The highly oriented mesopores with a size of ~3.2 nm along the c-axis can be well identified in TEM image (Figure 1a inset), in good agreement with the size distribution analysis above (Figure S2). Upon pyrolysis at 800 °C and subsequent ZrO₂ removal, Fe₂₀-PCN-222 can be converted to the Fe_{SA}-N-C without any identifiable metallic particle, and the rod shape and highly oriented mesopores in Fe₂₀-PCN-222 are inherited to the resultant Fe_{SA}-N-C to a large extent (Figure 1b, c). N₂ sorption for Fe_{SA}-N-C suggests its high BET surface area of 532 m²/g and the evident hysteresis loop (P/P₀: 0.6-0.8) indicates its mesoporous structure (Figure S4a), which is recognizable in the cross-section SEM and TEM images (Figure 1c-e and S3b). The pore size distribution analysis further manifests its hierarchical pores in 1-2 nm and ~9 nm diameters (Figure S4b). It is generally believed that micropores would increase the density of active sites and mesopores are beneficial to the mass transfer, thus improving catalytic activity.^{5d,9b}

Powder X-ray diffraction (PXRD) pattern of Fe_{SA}-N-C presents two broad diffraction peaks in the range of 20-30° and 40-45°, respectively corresponding to the (002) and (101) reflections of graphitized carbon, and no peak assignable to iron-based species can be detected (Figure S5). Raman scattering spectrum for Fe_{SA}-N-C gives low intensity ratio (I_D/I_G = 0.95) of D band (~1345 cm⁻¹) and G band (~1590 cm⁻¹), further indicating its high graphitization degree, which favors the electron transfer (Figure S6). X-ray photoelectron spectroscopy (XPS) results reveal that C, N, O and Fe elements are included in Fe_{SA}-N-C (Figure S7a). High-resolution N 1s spectrum can be fitted into five subpeaks, pyridinic N (398.5 eV), Fe-N_x (399.2 eV), pyrrolic N (400.3 eV), graphitic N (401.2 eV) and oxidized N (402.7 eV), in which the Fe-N_x peak suggests the evolution of Fe-N_x units after pyrolysis (Figure 2a). High-resolution Fe 2p spectrum with two relatively weak peaks centered at 710.7 eV (Fe 2p_{3/2}) and 723.1 eV (Fe 2p_{1/2}) illustrates partially oxidized Fe species in Fe_{SA}-N-C, further supporting the existence of Fe-N_x species (Figure S7b).^{8c} The existence of abundant N species (4.67 wt%) and a high loading of Fe (up to 1.76 wt%), perfectly consistent with the XPS results, have been quantified by elemental analysis and inductively coupled plasma atomic emission spectrometry (ICP-AES) (Table S1). Aberration-corrected high-angle annular dark field scanning transmission electron microscope (HAADF-STEM) observation unambiguously identifies single Fe atoms from the bright spots in randomly selected areas, but no Fe particle is found (Figure 1f and S8).

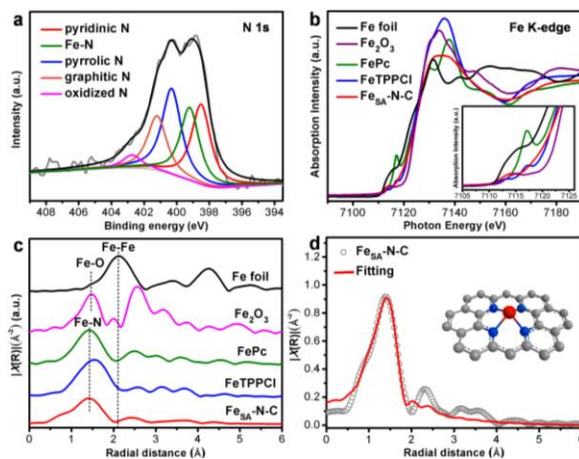


Figure 2. (a) High-resolution XPS spectrum of N 1s for FeSA-N-C. (b) Fe K-edge X-ray absorption near-edge structure (XANES) and (c) Fourier transform-extended X-ray absorption fine structure (FT-EXAFS) spectra of FeSA-N-C, 5,10,15,20-tetraphenylporphineiron(III) chloride (FeTPPCL) and Iron Phthalocyanine (FePc), Fe₂O₃ and Fe foil. (d) EXAFS fitting for FeSA-N-C (inset: schematic model of FeSA-N-C. The red, blue and grey spheres represent Fe, N and C atoms, respectively).

In addition to the localized microstructural information above, X-ray absorption spectroscopy (XAS) was further studied to unveil the electronic and structural information of Fe-involved species in FeSA-N-C. From the Fe K-edge XANES spectra, the energy absorption threshold of FeSA-N-C locates between Fe foil and Fe₂O₃ that is more close to that of FeTPPCL and FePc, implying the positively charged Fe^{δ+} sites stabilized by N atoms in FeSA-N-C (Figure 2b). Furthermore, the absence of the edge peak at ~7117.1 eV, which is the fingerprint of square-planar D_{4h} symmetry, suggests lower symmetry of Fe-N_x sites in FeSA-N-C with a non-planar configuration.^{3b} FT-EXAFS spectrum of FeSA-N-C at Fe K-edge presents only a main peak at ~1.44 Å, which is attributed to the Fe-N scattering path, and no Fe-Fe bond at ~2.13 Å can be detected, manifesting that the evolution of single-atom Fe sites in FeSA-N-C (Figure 2c). To further verify the coordination structure around Fe centers, EXAFS fitting has been performed (Figure 2d, S9 and Table S2). The best fitting result for the first shell shows that each Fe atom is coordinated by ~4 N atoms in average, illustrating that all atomic Fe sites are four-coordinated by nitrogen species (Figure 2d inset). It is noteworthy that when the molar percentage of Fe-TCPP increases to 40% in PCN-222, Fe-Fe peak is identifiable in the resultant catalyst (denoted as Fe_{NP}-N-C), manifesting that Fe atoms might aggregate to nanoparticles (NPs) upon pyrolysis with a higher percentage of Fe-TCPP, in consistent with high-resolution TEM observation results (Figure S10).

The above results prove that precursor Fe₂₀-PCN-222 might possess an appropriate Fe-TCPP content (20 mol%) to prepare single Fe atoms. The fabrication mechanism for single Fe atoms and Fe NPs has been proposed (Scheme S3). Firstly, the mixed ligands of H₂-TCPP and Fe-TCPP (Fe is coordinated by N atoms) are employed to construct PCN-222. Its 3D skeleton composed by metal nodes and organic ligands creates the isolation of porphyrin ligands. It is convenient to modulate the average distance (d) of adjacent Fe-TCPP ligands by simply changing the ratio of the mixed ligands. When the distance reaches to a critical value (for example, in Fe₂₀-PCN-222), the formation of Fe NPs is effectively inhibited in pyrolysis process. Upon the removal of ZrO₂, single Fe atoms anchored N-doped carbon can be successfully obtained (Scheme S3b). Conversely, when Fe-TCPP loading becomes higher and Fe...Fe distance is shortened, Fe aggregation would be inevitable during the MOF pyrolysis (Scheme S3c).

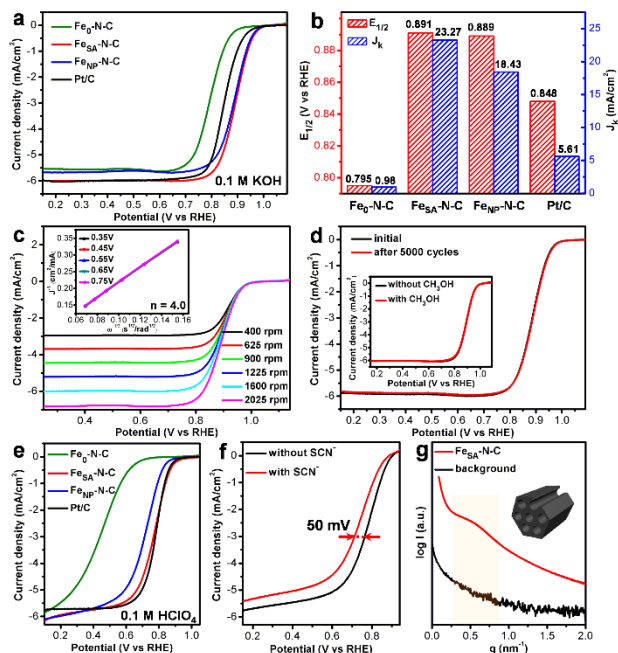


Figure 3. (a) LSV curves and (b) $E_{1/2}$ and J_k at 0.85 V for various catalysts in 0.1 M KOH. (c) LSV curves of Fe_{SA}-N-C at different rotating rates in 0.1 M KOH (inset in c: K-L plots and electron transfer number). (d) LSV curves of Fe_{SA}-N-C before and after 5000 cycles (inset: LSV curves before and after the addition of CH₃OH) in 0.1 M KOH. (e) LSV curves of various catalysts in 0.1 M HClO₄. (f) LSV curves of Fe_{SA}-N-C in 0.1 M HClO₄ before and after the addition of SCN⁻. (g) SAXS pattern of Fe_{SA}-N-C.

In view of the intrinsic activity of Fe-N₄ sites for efficient oxygen reduction reaction (ORR),³ the performance of Fe_{SA}-N-C was first investigated in 0.1 M KOH solution. For Fe₀-N-C sample (obtained from Fe₀-PCN-222) without Fe doping, its CV curve displays a much lower oxygen reduction peak than that of Fe_{SA}-N-C and Fe_{NP}-N-C, reflecting an indispensable role of Fe-N sites on boosting ORR activity (Figure S11). Linear sweep voltammetry (LSV) curves indicate the best performance of Fe_{SA}-N-C among all related catalysts, featuring a higher half-wave potential ($E_{1/2}$ = 0.891 V) than Fe₀-N-C (0.795 V), Fe_{NP}-N-C (0.889 V), and the state-of-the-art Pt/C catalyst (0.848 V) (Figure 3a, b). In addition, the superior activity of Fe_{SA}-N-C is supported by the far superior kinetic current density (J_k) at 0.85 V (23.27 mA/cm²) to that of Fe₀-N-C (0.98 mA/cm²), Fe_{NP}-N-C (18.43 mA/cm²) and commercial Pt/C (5.61 mA/cm²), signifying its much more favorable kinetics (Figure 3b).

To gain more insight into the electron-transfer mechanism of Fe_{SA}-N-C, LSV curves at different rotating rates of rotating disk electrode (RDE) were recorded (Figure 3c). The K-L plots obtained from the LSV curves exhibit excellent linearity, presenting the first order reaction kinetics for ORR with a potential-independent electron transfer rate (Figure 3c inset).¹¹ Based on the K-L equation, the electron transfer number is determined to be ~4.0, manifesting an ideal 4e⁻ ORR mechanism. Rotating ring disk electrode (RRDE) measurement for Fe_{SA}-N-C was further performed to show the very low (<5.5%) H₂O₂ yields with an electron transfer number larger than 3.9 (Figure S12), similar to the results from the K-L plots mentioned above. In addition, the negligible decay of LSV curves after continuous 5000 cycles and the methanol addition experiment for Fe_{SA}-N-C demonstrate its excellent durability and strong tolerance against the methanol-crossover effect, in stark contrast to the significant decline of Pt/C (Figure 3d and S13). The results suggest that Fe_{SA}-N-C possesses the best ORR performance among all non-noble metal catalysts ever reported under alkaline condition (Figure 3a-d and Table S3).

The excellent ORR performance of Fe_{SA}-N-C in alkaline media prompted our further exploration under more challenging acidic condition. Strikingly, superior ORR activity with a $E_{1/2}$ of 0.776 V in 0.1 M HClO₄ can be achieved for Fe_{SA}-N-C, which is much higher than that of Fe₀-N-C and Fe_{NP}-N-C and even comparable to (5 mV lower) that of benchmark Pt/C (Figure 3e and S14a,b). The very low kinetic current density (9.60 mA/cm²) at 0.75 V manifests the favorable kinetics of Fe_{SA}-N-C (Figure S14b). Further RRDE test suggests a H₂O₂ yield lower than 1.0% with an ideal 4e⁻ transfer process, in consistency with the result from K-L plots at different rotation rates (Figure S14c, d). Moreover, Fe_{SA}-N-C exhibits an excellent durability with only 6 mV decay after 5000 cycles, which is much better than the Pt/C, possibly due to the strong affinity between atomic Fe sites and coordination N within carbon (Figure S15a,c). Additionally, Fe_{SA}-N-C demonstrates a superior tolerance to the crossover effect of methanol compared with Pt/C, making it a promising candidate for the direct application of methanol fuel cells (Figure S15b,d). It is noteworthy that, although much endeavor has been devoted, very limited catalysts were reported simultaneously possessing comparable performance with Pt/C in both alkaline and acidic media, further highlighting the huge superiority of the single Fe atom-involved Fe_{SA}-N-C over other reported catalysts (Table S4).

To further understand the origin of the above excellent ORR performance, more investigations have been carried out on Fe_{SA}-N-C catalyst. The SCN⁻ (0.01 M), with strong affinity to Fe ion, was employed as a probe to poison Fe-N₄ sites.^{9c} As a result, the half-wave potential of Fe_{SA}-N-C for ORR in 0.1 M HClO₄ decreased significantly by 50 mV (Figure 3f). The SCN⁻-poisoned electrode was subsequently rinsed with pure water and re-measured in 0.1 M KOH. The $E_{1/2}$ of the first LSV curve negatively shifts by 10 mV compared with original catalyst. With increasing cycle numbers, the LSV curves gradually overlap with the original one obtained before SCN⁻ treatment, which should be attributed to the desorption of SCN⁻ from Fe site in 0.1 M KOH (Figure S16). The activity recovery accompanied by the release of SCN⁻ from blocked Fe-N₄ sites clearly reflects that single Fe atoms stabilized by Fe-N interaction are responsible for the high ORR activity of Fe_{SA}-N-C.

In addition, the hierarchically porous structure in Fe_{SA}-N-C is generally believed to affect the accessibility of active sites and mass transfer in catalytic process. Small angle X-ray scattering (SAXS) for Fe_{SA}-C-N gives broad hump between 0.28 to 0.84 nm⁻¹ of Fe_{SA}-C-N, hinting the ordering of pore structure with size distribution from 7 to 22 nm (Figure 3g), in good agreement with the above microstructure observation and N₂ sorption data (Figure 1d,e and S4). It should be pointed out that, though reported MOF-derived carbon materials exhibit high surface areas, their pore structures are usually disordered and non-interconnected after uncontrollable pyrolysis, which is not favorable to mass transfer in catalysis.^{7e,8e} Therefore, the development of porous carbon with oriented channels, which remains to be a long-term target yet a great challenge to date, has been achieved for the first time. The hierarchically porous Fe_{SA}-N-C, possessing abundant micropores (accommodating high-density Fe-N₄ active sites) and open mesoporous channels with particular orientation (facilitating the high-flux mass transfer), would greatly accelerate the ORR process (Figure S17). This is in sharp contrast to the sluggish mass transfer in traditional MOF-derived carbons with non-interconnected pores. Taking the above Fe-N₄ active sites and pore structure merits together, we may come to the conclusion that single-atom Fe sites, accessible by the hierarchal pores in Fe_{SA}-N-C mainly contribute to the superb ORR performance.

In summary, we have come up with a novel mixed-ligand strategy in MOF system to fabricate single-atom catalyst for the first time. The ratio optimization of the mixed porphyrin ligands with and without Fe^{III} centers gives rise to long spatial distance of Fe^{III} ions in the MOF skeleton, favoring the formation of single Fe atoms upon pyrolysis. Benefiting from periodic and tailorable MOF structures, the spatial distance control of Fe^{III} in porphyrinic MOFs effectively suppresses the Fe aggregation during pyrolysis, leading to atomic dispersion of Fe with a high loading (1.76 wt%) in the resultant porous carbon. Thanks to the single-atom Fe sites (superior activity), hierarchically porous structure (accessible active sites) with oriented mesopores (fast diffusion of O₂ and intermediates) and high conductivity (fast electron transfer), Fe_{SA}-N-C demonstrates excellent ORR performance in both alkaline and acidic media, surpassing any other reported non-noble-metal catalysts and even the state-of-the-art Pt/C. In view of the huge diversity and tailorability of MOFs, this work opens up a new avenue to the rational synthesis of efficient SACs via multiscale control and related research along this line is ongoing in our laboratory.

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